FINITENESS OF A BASE-CHANGED ALGEBRA MAP IS REALISED AT A FINITE STAGE OF THE COLIMIT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Finiteness of an algebra map via monic polynomial relations. Used at the start of the counted proof (lines 47753-47755). Produces finitely many generators and monic polynomials witnessing finiteness after base change.
- Tensor products commute with directed colimits. Used at the end of the counted proof (lines 47757-47759). Lets the finitely many polynomial data be moved down to a single stage of the colimit.

Lemma 0.1. *Let $A = \operatorname{colim}_{i \in I} A_i$ be a directed colimit of rings. Let $0 \in I$ and $\varphi_0 : B_0 \to C_0$ a map of A_0-algebras. Assume*

(1) *$A \otimes_{A_0} B_0 \to A \otimes_{A_0} C_0$ is finite,*
(2) *C_0 is of finite type over B_0.*

Then there exists an $i \geq 0$ such that the map $A_i \otimes_{A_0} B_0 \to A_i \otimes_{A_0} C_0$ is finite.

Proof. Let $x_1, \ldots, x_m$ be generators for C_0 over B_0. Pick monic polynomials $P_j \in A \otimes_{A_0} B_0[T]$ such that $P_j(1 \otimes x_j) = 0$ in $A \otimes_{A_0} C_0$. For some $i \geq 0$ we can find $P_{j,i} \in A_i \otimes_{A_0} B_0[T]$ mapping to P_j. Since $\otimes$ commutes with colimits we see that $P_{j,i}(1 \otimes x_j)$ is zero in $A_i \otimes_{A_0} C_0$ after possibly increasing i. Then this i works. □

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